

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402298
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Lai; Jen-I

Semiconductor structure including bit line structure covered by oxide barrier layer and manufacturing method thereof

Abstract

A semiconductor structure includes a substrate, a bit line structure, an oxide barrier layer, and a bit line capping layer. The bit line structure is disposed on the substrate, in which the bit line structure includes a conductive silicon layer, a conductive layer, and a hard mask layer. The conductive layer is disposed on the conductive silicon layer, in which the conductive layer includes a metal. The hard mask layer is disposed on the conductive layer. The oxide barrier layer is disposed in direct contact with a sidewall of the bit line structure. The bit line capping layer covers the oxide barrier layer.

Inventors:	Lai; Jen-I (Taoyuan, TW)
Applicant:	NANYA TECHNOLOGY CORPORATION (New Taipei, TW)
Family ID:	1000008777867
Assignee:	NANYA TECHNOLOGY CORPORATION (New Taipei, TW)
Appl. No.:	18/056724
Filed:	November 18, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240172414 A1	May. 23, 2024

Publication Classification

Int. Cl.:	H10B12/00 (20230101)
U.S. Cl.:	

Field of Classification Search

CPC: H10B (12/482); H10B (12/30)

USPC: 257/295

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6090701	12/1999	Hasunuma	438/650	H01L 21/76882
6550915	12/2002	Grobe, III	525/326.4	C08J 7/123
6746925	12/2003	Lin	438/452	H01L 21/32105
11991876	12/2023	Pan	N/A	H10B 12/053
2007/0007578	12/2006	Li	257/315	H01L 29/40114
2010/0267247	12/2009	Ma	438/770	H01L 21/32105
2014/0061806	12/2013	Eun	257/368	H10B 12/485
2020/0006533	12/2019	Tsai	N/A	H01L 29/785
2022/0052049	12/2021	Chen	N/A	H10B 12/30
2022/0223603	12/2021	Ping	N/A	H10B 12/482
2023/0067536	12/2022	Lu	N/A	H10B 12/34
2024/0008263	12/2023	Lu	N/A	H10B 12/482

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102446831	12/2011	CN	N/A
102544354	12/2011	CN	N/A
112018080	12/2019	CN	N/A
113035873	12/2020	CN	N/A
20050028961	12/2004	KR	N/A
20090037553	12/2008	KR	N/A
202137415	12/2020	TW	N/A
WO-9917351	12/1998	WO	H01L 21/18

Primary Examiner: Zhu; Sheng-Bai

Attorney, Agent or Firm: CKC & Partners Co., LLC

Background/Summary

BACKGROUND

Field of Invention

(1) The present disclosure relates to a semiconductor structure and a manufacturing method thereof.

Description of Related Art

(2) A dynamic random access memory (DRAM) is a semiconductor arrangement for storing bits of data with cell capacitors within an integrated circuit. DRAMs commonly include trench capacitor

DRAM cells and/or stacked capacitor DRAM cells.

(3) As DRAM devices become more highly integrated, design rules of the DRAM devices become finer. However, as the size of a DRAM device is reduced, the DRAM device may have a problem associated with leakage current. Therefore, there is need to use a capping layer to protect components in the DRAM device to be electrically insulated from other components. However, during the manufacturing process of the capping layer, the electrical performance of the components may be influenced. To overcome the performance issue, there is a significant need to improve the manufacturing process.

SUMMARY

(4) The present disclosure provides a semiconductor structure including a substrate, a bit line structure, an oxide barrier layer, and a bit line capping layer. The bit line structure is disposed on the substrate, in which the bit line structure includes a conductive silicon layer, a conductive layer, and a hard mask layer. The conductive layer is disposed on the conductive silicon layer, in which the conductive layer includes a first metal. The hard mask layer is disposed on the conductive layer. The oxide barrier layer is disposed in direct contact with a first sidewall of the bit line structure. The bit line capping layer covers the oxide barrier layer.

(5) In some embodiments, the oxide barrier layer includes a first portion in direct contact with the conductive silicon layer and a second portion in direct contact with the conductive layer, the first portion includes SiO, SiO.sub.2, or a combination thereof, and the second portion includes a first oxide of the first metal.

(6) In some embodiments, the oxide barrier layer further includes a third portion in direct contact with the hard mask layer, the hard mask layer includes an insulating material selected from the group consisting of Si.sub.3N.sub.4, SiCN, SiC, and SiO.sub.2, and the third portion includes a second oxide of the insulating material of the hard mask layer.

(7) In some embodiments, the semiconductor structure further includes a first barrier layer disposed between the conductive silicon layer and the conductive layer, in which the first barrier layer includes a second metal, the oxide barrier layer further includes a third portion in direct contact with the first barrier layer, and the third portion includes a second oxide of the second metal.

(8) In some embodiments, the first barrier layer includes a metal layer, a metal nitride layer, a metal silicide layer, or combinations thereof.

(9) In some embodiments, the metal layer includes a metal selected from the group consisting of Co, Cu, Ni, Ru, Mn, Ag, Au, Pt, Fe, Mo, Rh, Ti, Ta, and W, the metal nitride layer includes a material selected from the group consisting of tungsten nitride, TiN, and TaN, and the metal silicide layer includes a material selected from the group consisting of tungsten silicide, tantalum silicide, titanium silicide, molybdenum silicide, zirconium silicide, cobalt silicide, chromium silicide, and nickel silicide.

(10) In some embodiments, the semiconductor structure further includes a second barrier layer disposed between the conductive silicon layer and the first barrier layer, in which the second barrier layer includes a third metal, the oxide barrier layer further includes a fourth portion in direct contact with the second barrier layer, and the fourth portion includes a third oxide of the third metal.

(11) In some embodiments, the first barrier layer is a first metal nitride layer or a metal silicide layer, the second barrier layer is a second metal nitride layer or a metal layer, and the first metal nitride layer and the second metal nitride layer have different materials.

(12) In some embodiments, the semiconductor structure further includes a third barrier layer disposed between the first barrier layer and the conductive layer, in which the third barrier layer includes a fourth metal, the oxide barrier layer further includes a fifth portion in direct contact with the third barrier layer, and the fifth portion includes a fourth oxide of the fourth metal.

(13) In some embodiments, the first barrier layer is a metal silicide layer, the second barrier layer is a metal layer or a first metal nitride layer, the third barrier layer is a second metal nitride layer, and

the first metal nitride layer and the second metal nitride layer have different materials.

(14) In some embodiments, the bit line capping layer includes a first nitride layer, an oxide layer, and a second nitride layer. The first nitride layer covers a second sidewall of the oxide barrier layer. The oxide layer covers a third sidewall of the first nitride layer. The second nitride layer covers a fourth sidewall of the oxide layer and a top surface of the bit line structure.

(15) In some embodiments, the oxide barrier layer consists of SiO₂.

(16) In some embodiments, the conductive layer includes a metal selected from the group consisting of W, Ru, Ir, Pt, Rh, and Mo.

(17) The present disclosure provides a method of manufacturing a semiconductor structure including the following operations. A bit line structure disposed on a substrate is received, in which the bit line structure includes a conductive silicon layer, a conductive layer, and a hard mask layer. The conductive layer is disposed on the conductive silicon layer, in which the conductive layer includes a first metal. The hard mask layer is disposed on the conductive layer. An oxide barrier layer is formed in direct contact with a sidewall of the bit line structure. A bit line capping layer is formed to cover the oxide barrier layer.

(18) In some embodiments, forming the oxide barrier layer in direct contact with the sidewall of the bit line structure includes: performing a plasma treatment on the bit line structure to form the oxide barrier layer by oxidizing a first surface portion of the conductive silicon layer, a second surface portion of the conductive layer, and a third surface portion of the hard mask layer.

(19) In some embodiments, the plasma treatment is performed at a temperature between 225° C. and 275° C., and a plasma gas is generated from an oxidizing gas and a reducing gas.

(20) In some embodiments, a flow rate percentage of the reducing gas is between 30% and 50% in the plasma gas.

(21) In some embodiments, a treatment time of the plasma treatment is between 20 seconds and 45 seconds.

(22) In some embodiments, the bit line structure further includes: a first barrier layer disposed between the conductive silicon layer and the conductive layer, in which the first barrier layer includes a second metal.

(23) In some embodiments, forming the oxide barrier layer in direct contact with the sidewall of the bit line structure includes: performing a plasma treatment on the bit line structure to form the oxide barrier layer by oxidizing a first surface portion of the conductive silicon layer, a second surface portion of the conductive layer, a third surface portion of the hard mask layer, and a fourth surface portion of the first barrier layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

(2) FIGS. 1A-1F are cross-sectional views schematically illustrating intermediate stages in the manufacturing of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(3) FIGS. 2-5 are cross-sectional views of semiconductor structures, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(4) Reference will now be made in detail to the embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

(5) The following embodiments are disclosed with accompanying diagrams for detailed

description. For illustration clarity, many details of practice are explained in the following descriptions. However, it should be understood that these details of practice do not intend to limit the present disclosure. That is, these details of practice are not necessary in parts of embodiments of the present disclosure. Furthermore, for simplifying the drawings, some of the conventional structures and elements are shown with schematic illustrations.

(6) The present disclosure provides a method of manufacturing a semiconductor structure. The semiconductor structure includes a bit line structure, an oxide barrier layer, and a bit line capping layer. During the manufacturing process, the oxide barrier layer is formed on the bit line structure, and subsequently the bit line capping layer is formed on the oxide barrier layer. Since the bit line structure is protected by the oxide barrier layer, the forming step of the bit line capping layer may not adversely influence the bit line structure. For example, the element of the bit line capping layer may not diffuse into the bit line structure to influence the electrical performance of the bit line structure. In other words, the oxide barrier layer can block diffusion element migration from the bit line capping layer. Furthermore, the bit line structure of the present disclosure may be disposed in a DRAM device.

(7) The present disclosure provides several embodiments for further illustration. FIGS. 1A-1F are cross-sectional views schematically illustrating intermediate stages in the manufacturing of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(8) Reference is made to FIG. 1A. A bit line structure **120** disposed on a substrate **110** is received, in which the bit line structure **120** includes a conductive silicon layer **122**, a conductive layer **124**, and a hard mask layer **126**. The conductive silicon layer **122** is disposed on the substrate **110**. The conductive layer **124** is disposed on the conductive silicon layer **122**, in which the conductive layer **124** includes a first metal. The hard mask layer **126** is disposed on the conductive layer **124**. As shown in FIG. 1A, the conductive silicon layer **122** directly contacts with the conductive layer **124**. However, in other embodiments, one or more barrier layers (not shown) are disposed between the conductive silicon layer **122** and the conductive layer **124**. The conductive layer **124** directly contacts with the hard mask layer **126**.

(9) Please still referring to FIG. 1A, in some embodiments, the substrate **110** includes an understructure (not shown) including landing plugs (not shown) to be electrically connected with a bit line structure. In some embodiments, the substrate **110** is one of a single crystal silicon wafer, a silicon on insulator (SOI) substrate, an epitaxial growth layer, or other semiconducting layers. In some embodiments, the conductive silicon layer **122** includes polysilicon. In some embodiments, the conductive layer **124** includes a metal selected from the group consisting of W, Ru, Ir, Pt, Rh, and Mo. In some embodiments, the hard mask layer **126** includes an insulating material selected from the group consisting of Si.sub.3N.sub.4, SiCN, SiC, and SiO.sub.2.

(10) Attention is now invited to FIG. 1B. An oxide barrier layer **130** is formed in direct contact with a first sidewall SW1 and a top surface T of the bit line structure **120**. In some embodiments, the oxide barrier layer **130** is formed by performing a plasma treatment on the bit line structure **120** to form the oxide barrier layer **130** by oxidizing a first surface portion of the conductive silicon layer **122**, a second surface portion of the conductive layer **124**, and a third surface portion of the hard mask layer **126**. Therefore, the oxide barrier layer **130** includes a first portion **132** in direct contact with the conductive silicon layer **122**, a second portion **134** in direct contact with the conductive layer **124**, and a third portion **136** in direct contact with the hard mask layer **126**. Since the conductive silicon layer **122** includes silicon, the first portion **132** of the oxide barrier layer **130** includes SiO, SiO.sub.2, or a combination thereof. Since the conductive layer **124** includes a first metal, the second portion **134** of the oxide barrier layer **130** includes a first oxide of the first metal. In some embodiments, the first oxide of the first metal includes an oxide of W, Ru, Ir, Pt, Rh, Mo, or alloy of these metals. In some embodiments, since the hard mask layer **126** includes an insulating material selected from the group consisting of Si.sub.3N.sub.4, SiCN, SiC, and SiO.sub.2, the third portion **136** of the oxide barrier layer **130** includes a second oxide of the

insulating material of the hard mask layer **126**.

(11) In some embodiments, the plasma treatment is performed at a temperature between 225° C. and 275° C., and a plasma gas is generated from an oxidizing gas and a reducing gas. The oxide barrier layer **130** can have a proper thickness under this temperature range. In some embodiments, the plasma treatment is performed in dry strip equipment. During the plasma treatment, surface portions of the bit line structure **120** are oxidized by plasma generated from the oxidizing gas to form the oxide barrier layer **130**. Simultaneously, a portion of the oxide barrier layer **130** is reduced by plasma generated from the reducing gas. The reducing gas may prevent an excess of the bit line structure **120** from being oxidized, and therefore the resistance of the bit line structure **120** may not be greatly influenced and/or increased. Simultaneous use of the oxidizing gas and the reducing gas can form the oxide barrier layer **130** with a proper thickness. Therefore, the oxide barrier layer **130** can have good ability to block element diffusion into the bit line structure **120** during forming a bit line capping layer on the oxide barrier layer **130** (described in FIGS. **1C-1E**) without affecting the electrical performance of the bit line structure **120**. In some embodiments, the oxide barrier layer **130** has a thickness between 3 angstrom and 5 angstrom. In some embodiments, the oxide barrier layer **130** has a substantially uniform thickness. In some embodiments, a treatment time of the plasma treatment is between 20 seconds and 45 seconds. In some embodiments, a plasma power of the plasma treatment is between 3500 W and 4500 W, such as 3500, 3700, 3900, 4100, 4300, or 4500 W.

(12) For example, the oxidizing gas includes O.sub.2, and the reducing gas includes NH.sub.3. In some embodiments, a flow rate percentage of the reducing gas is between 30% and 50% in the plasma gas. For example, a flow rate percentage of NH.sub.3 is between 30% and 50% in the plasma gas containing O.sub.2 and NH.sub.3. If the flow rate percentage is greater than 50%, the oxide barrier layer **130** may not have enough thickness to block element diffusion. If flow rate percentage is less than 30%, an excess of the bit line structure **120** may be oxidized, and its electrical performance may be influenced.

(13) Please still refer to FIG. **1B**. In some embodiments, the conductive silicon layer **122** includes polysilicon, the conductive layer **124** includes W, and the hard mask layer **126** includes Si.sub.3N.sub.4. The oxide barrier layer **130** is formed by oxidizing surface portions of the conductive silicon layer **122**, the conductive layer **124**, and the hard mask layer **126**. The oxide barrier layer **130** includes the first portion **132** containing SiO, SiO.sub.2, or a combination thereof, the second portion **134** containing WO.sub.3, and the third portion **136** containing SiON.

(14) In some embodiments, before forming the oxide barrier layer **130**, the bit line structure **120** is etch cleaned by, for example, dilute hydrofluoric acid (DHF) and NH.sub.4OH, to remove the remains on the surface of the bit line structure **120**, which is byproducts of patterning the bit line structure **120** by, for example, dry etch.

(15) Reference is made to FIGS. **1C-1E**. A bit line capping layer BC is formed to cover the bit line structure **120** and the oxide barrier layer **130**. In some embodiments, the bit line capping layer BC is a single layer or multilayers. The single layer may be an insulating nitride layer. In some embodiments, the bit line capping layer BC is called a bit line spacer and is an insulating layer.

(16) As shown in FIG. **1C**, a first nitride layer **140** is formed to cover a second sidewall SW2 and the top surface of the oxide barrier layer **130** and the substrate **110**. In some embodiments, the first nitride layer **140** is in direct contact with the first portion **132**, the second portion **134**, and the third portion **136** of the oxide barrier layer **130**. In some embodiments, the first nitride layer **140** is an insulating layer. In some embodiments, the first nitride layer **140** includes an insulating material selected from the group consisting of Si.sub.3N.sub.4 and SiCN. In some embodiments, the first nitride layer **140** is formed by an atomic layer deposition (ALD) or a chemical vapor deposition (CVD). Since the bit line structure **120** is protected by the oxide barrier layer **130**, the forming step of the first nitride layer **140** may not adversely influence the bit line structure **120**. For example, the element of the first nitride layer **140**, such nitrogen (N), may not diffuse or penetrate into the bit

line structure **120** to influence the electrical performance of the bit line structure **120**. The oxide barrier layer **130** can block diffusion nitrogen migration. For example, the conductive layer **124** may include tungsten (W). If the conductive layer **124** is not protected by the oxide barrier layer **130**, tungsten may react with the nitrogen of the first nitride layer **140** to form tungsten nitride, which is high k material. In other words, sidewall portions of the conductive layer **124** include tungsten nitride that abnormally grows. Therefore, the overall electrical performance of the bit line structure **120** may be adversely influenced, the resistance of the conductive layer **124** increases, and therefore its electrical performance cannot satisfy the design requirement.

(17) Attention is now invited to FIGS. **1C-1D**. Portions of the first nitride layer **140** that are in direct contact with the substrate **110** and the top surface of the oxide barrier layer **130** are removed, and therefore the remaining first nitride layer **140** are disposed adjacent two sides of the bit line structure **120** and two sides of the oxide barrier layer **130** as shown in FIG. **1D**. Moreover, an oxide layer **150** is formed to cover a third sidewall SW3 of the first nitride layer **140**, the substrate **110**, and the oxide barrier layer **130**. In some embodiments, the oxide layer **150** is in direct contact with the top surface of the third portion **136** of the oxide barrier layer **130**. In some embodiments, the oxide layer **150** is an insulating layer. In some embodiments, the oxide layer **150** includes SiO.sub.2.

(18) Please refer to FIGS. **1D-1E**. Portions of the oxide layer **150** that are in direct contact with the substrate **110**, the top surface of the oxide barrier layer **130**, and the top surface of the first nitride layer **140** are removed, and therefore the remaining oxide layer **150** are disposed adjacent two sides of the bit line structure **120**, two sides of the oxide barrier layer **130**, and two sides of the first nitride layer **140** as shown in FIG. **1E**. Moreover, a second nitride layer **160** is formed to cover a fourth sidewall SW4 of the oxide layer **150**, the top surface T of the bit line structure **120**, the oxide barrier layer **130**, and the first nitride layer **140**. Therefore, the bit line capping layer BC including the first nitride layer **140**, the oxide layer **150**, and the second nitride layer **160** is formed. In some embodiments, the second nitride layer **160** includes an insulating material selected from the group consisting of Si.sub.3N.sub.4 and SiCN. In some embodiments, the second nitride layer **160** is formed by an atomic layer deposition (ALD) or a chemical vapor deposition (CVD).

(19) Reference is made to FIG. **1F**. An interlayer dielectric layer ILD is formed to cover the bit line capping layer BC. As shown in FIG. **1F**, the bit line structure **120** is disposed on the substrate **110** and includes the conductive silicon layer **122**, the conductive layer **124**, and the hard mask layer **126**. The conductive layer **124** includes a first metal. The oxide barrier layer **130** covers the bit line structure **120** and includes the first portion **132** containing SiO, SiO.sub.2, or a combination thereof, the second portion **134** containing the first oxide of the first metal, and the third portion **136** containing the second oxide of the insulating material of the hard mask layer **126**. The bit line capping layer BC including the first nitride layer **140**, the oxide layer **150**, and the second nitride layer **160** covers the oxide barrier layer **130**. It is noted that the oxide barrier layer **130** can protect the conductive components in the bit line structure **120**, i.e., the conductive silicon layer **122** and the conductive layer **124**, from being affected by nitrogen diffusion when forming the first nitride layer **140** and the second nitride layer **160**. Therefore, the semiconductor structure **100** can have excellent electrical performance.

(20) Next, please refer to FIGS. **2-5**. FIGS. **2-5** are cross-sectional views of semiconductor structures, in accordance with some embodiments of the present disclosure.

(21) Attention is now invited to FIG. **2**. A semiconductor structure **200** includes a substrate **110**, a bit line structure **120**, an oxide barrier layer **230**, a bit line capping layer BC, and an interlayer dielectric layer ILD. In some embodiments, the oxide barrier layer **230** is formed by an atomic layer deposition (ALD) or a chemical vapor deposition (CVD). In some embodiments, the oxide barrier layer **130** has a thickness between 3 angstrom and 5 angstrom. In some embodiments, the entire oxide barrier layer **230** comprises the same material, such as insulating oxide. For example, the oxide barrier layer **130** consists of SiO.sub.2. The difference between the semiconductor

structure **200** and the semiconductor structure **100** is that the oxide barrier layer **130** comprises three portions with different materials; however, the oxide barrier layer **230** comprises the same material.

(22) It is noted that the oxide barrier layer **230** can protect the conductive components in the bit line structure **120**, i.e., the conductive silicon layer **122** and the conductive layer **124**, from being affected by nitrogen diffusion when forming the first nitride layer **140** and the second nitride layer **160**. Therefore, the semiconductor structure **200** can have excellent electrical performance.

(23) Attention is now invited to FIG. 3. A semiconductor structure **300** includes a substrate **110**, a bit line structure **220**, an oxide barrier layer **330**, a bit line capping layer BC, and an interlayer dielectric layer ILD. The bit line structure **220** includes a conductive silicon layer **122**, a first barrier layer **322**, a conductive layer **124**, and a hard mask layer **126**. The difference between the semiconductor structure **300** and the semiconductor structure **100** is that the semiconductor structure **300** further includes the first barrier layer **322**, and the oxide barrier layer **330** further includes a fourth portion **332**. The first barrier layer **322** is disposed between the conductive silicon layer **122** and the conductive layer **124**, in which the first barrier layer **322** includes a second metal. In some embodiments, the oxide barrier layer **130** is formed by performing a plasma treatment on the bit line structure **220** to form the oxide barrier layer **130** by oxidizing a first surface portion of the conductive silicon layer **122**, a second surface portion of the conductive layer **124**, a third surface portion of the hard mask layer **126**, and a fourth surface portion of the first barrier layer **322**. Therefore, the oxide barrier layer **130** includes a first portion **132** in direct contact with the conductive silicon layer **122**, a second portion **134** in direct contact with the conductive layer **124**, a third portion **136** in direct contact with the hard mask layer **126**, and a fourth portion **332** in direct contact with the first barrier layer **322**. The fourth portion **332** includes a second oxide of the second metal. In some other embodiments, the oxide barrier layer **330** is replaced with the oxide barrier layer **230** shown in FIG. 2.

(24) Please still refer to FIG. 3. In some embodiments, the first barrier layer **322** includes a metal layer, a metal nitride layer, a metal silicide layer, or combinations thereof. In other words, the first barrier layer **322** includes a metal barrier layer, a metal nitride barrier layer, a metal silicide barrier layer, or combinations thereof. In some embodiments, the first barrier layer **322** includes two layers selected from the group consisting of the metal barrier layer, the metal nitride barrier layer, and the metal silicide barrier layer. In some embodiments, the first barrier layer **322** includes the metal barrier layer, the metal nitride barrier layer, and the metal silicide barrier layer.

(25) In some embodiments, the metal layer includes a metal selected from the group consisting of Co, Cu, Ni, Ru, Mn, Ag, Au, Pt, Fe, Mo, Rh, Ti, Ta, and W, the metal nitride layer includes a material selected from the group consisting of tungsten nitride, TiN, and TaN, and the metal silicide layer includes a material selected from the group consisting of tungsten silicide, tantalum silicide, titanium silicide, molybdenum silicide, zirconium silicide, cobalt silicide, chromium silicide, and nickel silicide. The tungsten nitride includes W.sub.2N, WN, WN.sub.2, or combinations thereof. The tungsten silicide includes WSi.sub.2, WSi, or a combination thereof.

(26) Please still refer to FIG. 3. In some embodiments, the conductive silicon layer **122** includes polysilicon, the first barrier layer **322** includes Ti, TiN, tungsten nitride, or tungsten silicide, the conductive layer **124** includes W, and the hard mask layer **126** includes Si.sub.3N.sub.4. The oxide barrier layer **330** is formed by oxidizing surface portions of the conductive silicon layer **122**, the first barrier layer **322**, the conductive layer **124**, and the hard mask layer **126**. The oxide barrier layer **130** includes the first portion **132** containing SiO, SiO.sub.2, or a combination thereof, the second portion **134** containing WO.sub.3, the third portion **136** containing SiON, and the fourth portion **332** containing TiO.sub.2 or WO.sub.3.

(27) Please refer to FIG. 4. A semiconductor structure **400** includes a substrate **110**, a bit line structure **420**, an oxide barrier layer **430**, a bit line capping layer BC, and an interlayer dielectric layer ILD. The bit line structure **420** includes a conductive silicon layer **122**, a first barrier layer

322, a second barrier layer **422**, a conductive layer **124**, and a hard mask layer **126**. The difference between the semiconductor structure **400** and the semiconductor structure **300** is that the semiconductor structure **400** further includes a second barrier layer **422**, and the oxide barrier layer **430** further includes a fifth portion **432**. The second barrier layer **422** is disposed between the conductive silicon layer **122** and the first barrier layer **322**, in which the second barrier layer **422** includes a third metal. In some embodiments, the oxide barrier layer **430** is formed by performing a plasma treatment on the bit line structure **420** to form the oxide barrier layer **430** by oxidizing a first surface portion of the conductive silicon layer **122**, a second surface portion of the conductive layer **124**, a third surface portion of the hard mask layer **126**, a fourth surface portion of the first barrier layer **322**, and a fifth surface portion of the second barrier layer **422**. Therefore, the oxide barrier layer **430** includes a first portion **132** in direct contact with the conductive silicon layer **122**, a second portion **134** in direct contact with the conductive layer **124**, a third portion **136** in direct contact with the hard mask layer **126**, a fourth portion **332** in direct contact with the first barrier layer **322**, and a fifth portion **432** in direct contact with the second barrier layer **422**. The fifth portion **432** includes a third oxide of the third metal. In some other embodiments, the oxide barrier layer **430** is replaced with the oxide barrier layer **230** shown in FIG. 2.

(28) Please still refer to FIG. 4. In some embodiments, the first barrier layer **322** is a first metal nitride layer or a metal silicide layer, the second barrier layer **422** is a second metal nitride layer or a metal layer, and the first metal nitride layer and the second metal nitride layer have different materials. In some embodiments, the first metal nitride layer and the second metal nitride layer independently include a material selected from the group consisting of tungsten nitride, TiN, and TaN. In some embodiments, the metal silicide layer includes a material selected from the group consisting of tungsten silicide, tantalum silicide, titanium silicide, molybdenum silicide, zirconium silicide, cobalt silicide, chromium silicide, and nickel silicide. In some embodiments, the metal layer includes a metal selected from the group consisting of Co, Cu, Ni, Ru, Mn, Ag, Au, Pt, Fe, Mo, Rh, Ti, Ta, and W.

(29) Please still refer to FIG. 4. In some embodiments, the conductive silicon layer **122** includes polysilicon, the first barrier layer **322** includes tungsten nitride or tungsten silicide, the second barrier layer **422** includes Ti or TiN, the conductive layer **124** includes W, and the hard mask layer **126** includes Si.sub.3N.sub.4. The oxide barrier layer **430** is formed by oxidizing surface portions of the conductive silicon layer **122**, the first barrier layer **322**, the second barrier layer **422**, the conductive layer **124**, and the hard mask layer **126**. The oxide barrier layer **430** includes the first portion **132** containing SiO, SiO.sub.2, or a combination thereof, the second portion **134** containing WO.sub.3, the third portion **136** containing SiON, and the fourth portion **332** containing WO.sub.3, and the fifth portion **432** containing TiO.sub.2.

(30) As shown in FIG. 5, a semiconductor structure **500** includes a substrate **110**, a bit line structure **520**, an oxide barrier layer **530**, a bit line capping layer BC, and an interlayer dielectric layer ILD. The bit line structure **520** includes a conductive silicon layer **122**, a first barrier layer **322**, a second barrier layer **422**, a third barrier layer **522**, a conductive layer **124**, and a hard mask layer **126**. The difference between the semiconductor structure **500** and the semiconductor structure **400** is that the semiconductor structure **500** further includes a third barrier layer **522**, and the oxide barrier layer **530** further includes a sixth portion **532**. The third barrier layer **522** is disposed between the first barrier layer **322** and the conductive layer **124**, in which the third barrier layer **522** includes a fourth metal. In some embodiments, the oxide barrier layer **530** is formed by performing a plasma treatment on the bit line structure **520** to form the oxide barrier layer **530** by oxidizing a first surface portion of the conductive silicon layer **122**, a second surface portion of the conductive layer **124**, a third surface portion of the hard mask layer **126**, a fourth surface portion of the first barrier layer **322**, a fifth surface portion of the second barrier layer **422**, and a sixth surface portion of the third barrier layer **522**. Therefore, the oxide barrier layer **530** includes a first portion **132** in direct contact with the conductive silicon layer **122**, a second portion **134** in direct contact with the

conductive layer **124**, a third portion **136** in direct contact with the hard mask layer **126**, a fourth portion **332** in direct contact with the first barrier layer **322**, a fifth portion **432** in direct contact with the second barrier layer **422**, and a sixth portion **532** in direct contact with the third barrier layer **522**. The sixth portion **532** includes a fourth oxide of the fourth metal. In some other embodiments, the oxide barrier layer **530** is replaced with the oxide barrier layer **230** shown in FIG. 2.

(31) Please still refer to FIG. 5. In some embodiments, the first barrier layer **322** is a metal silicide layer, the second barrier layer **422** is a metal layer or a first metal nitride layer, the third barrier layer **522** is a second metal nitride layer, and the first metal nitride layer and the second metal nitride layer have different materials. In some embodiments, the first metal nitride layer and the second metal nitride layer independently include a material selected from the group consisting of tungsten nitride, TiN, and TaN. In some embodiments, the metal silicide layer includes a material selected from the group consisting of tungsten silicide, tantalum silicide, titanium silicide, molybdenum silicide, zirconium silicide, cobalt silicide, chromium silicide, and nickel silicide. In some embodiments, the metal layer includes a metal selected from the group consisting of Co, Cu, Ni, Ru, Mn, Ag, Au, Pt, Fe, Mo, Rh, Ti, Ta, and W.

(32) Please still refer to FIG. 5. In some embodiments, the conductive silicon layer **122** includes polysilicon, the first barrier layer **322** includes tungsten silicide, the second barrier layer **422** includes Ti or TiN, the third barrier layer **522** includes tungsten nitride, the conductive layer **124** includes W, and the hard mask layer **126** includes Si.sub.3N.sub.4. The oxide barrier layer **530** is formed by oxidizing surface portions of the conductive silicon layer **122**, the first, second, and third barrier layers, **322**, **422**, **522**, the conductive layer **124**, and the hard mask layer **126**. The oxide barrier layer **530** includes the first portion **132** containing SiO, SiO.sub.2, or a combination thereof, the second portion **134** containing WO.sub.3, the third portion **136** containing SiON, and the fourth portion **332** containing WO.sub.3, the fifth portion **432** containing TiO.sub.2, and the sixth portion **532** containing WO.sub.3.

(33) The experiment conditions (temperature, plasma gas, flow rate percentage, treatment time, treatment time, plasma power) of forming the oxide barrier layers, **330**, **430**, **530**, and the thicknesses of the oxide barrier layers, **330**, **430**, **530**, can refer to the embodiments of the oxide barrier layer **130**, and the descriptions thereof will not be repeated.

(34) It is noted that the oxide barrier layers, **330**, **430**, **530**, can protect the conductive components in the bit line structures, i.e., the conductive silicon layer **122**, the conductive layer **124**, and barrier layers, from being affected by nitrogen diffusion when forming the first nitride layer **140** and the second nitride layer **160**. Therefore, the semiconductor structures, **300**, **400**, **500**, can have excellent electrical performance.

(35) In conclusion, the present disclosure provides a semiconductor structure and a manufacturing method thereof. The semiconductor structure includes a bit line structure covered by an oxide barrier layer that can block element diffusion into the bit line structure during forming a bit line capping layer on the oxide barrier layer. Therefore, the semiconductor structure can have excellent electrical performance.

(36) Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(37) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A semiconductor structure, comprising: a substrate; a bit line structure disposed on the substrate, wherein the bit line structure comprises: a conductive silicon layer; a conductive layer disposed on the conductive silicon layer, wherein the conductive layer comprises a first metal; and a hard mask layer disposed on the conductive layer, an oxide barrier layer disposed in direct contact with a first sidewall of the bit line structure and a top surface of the hard mask layer, wherein the oxide barrier layer comprises a first portion in direct contact with the conductive silicon layer, a second portion in direct contact with the conductive layer, and a third portion in direct contact with the hard mask layer, the first portion comprises SiO, SiO₂, or a combination thereof, the second portion comprises a first oxide of the first metal, the third portion comprises a second oxide of an insulating material of the hard mask layer, the insulating material of the hard mask layer is selected from the group consisting of Si₃N₄, SiCN, and SiC, and the first portion, the second portion, and the third portion of the oxide barrier layer are different materials; and a bit line capping layer covering the oxide barrier layer.
2. The semiconductor structure of claim 1, further comprising: a first barrier layer disposed between the conductive silicon layer and the conductive layer, wherein the first barrier layer comprises a second metal, the oxide barrier layer further comprises a fourth portion in direct contact with the first barrier layer, and the fourth portion comprises a third oxide of the second metal.
3. The semiconductor structure of claim 2, wherein the first barrier layer comprises a metal layer, a metal nitride layer, a metal silicide layer, or combinations thereof.
4. The semiconductor structure of claim 3, wherein the metal layer comprises a metal selected from the group consisting of Co, Cu, Ni, Ru, Mn, Ag, Au, Pt, Fe, Mo, Rh, Ti, Ta, and W, the metal nitride layer comprises a material selected from the group consisting of tungsten nitride, TiN, and TaN, and the metal silicide layer comprises a material selected from the group consisting of tungsten silicide, tantalum silicide, titanium silicide, molybdenum silicide, zirconium silicide, cobalt silicide, chromium silicide, and nickel silicide.
5. The semiconductor structure of claim 2, further comprising: a second barrier layer disposed between the conductive silicon layer and the first barrier layer, wherein the second barrier layer comprises a third metal, the oxide barrier layer further comprises a fifth portion in direct contact with the second barrier layer, and the fifth portion comprises a fourth oxide of the third metal.
6. The semiconductor structure of claim 5, wherein the first barrier layer is a first metal nitride layer or a metal silicide layer, the second barrier layer is a second metal nitride layer or a metal layer, and the first metal nitride layer and the second metal nitride layer have different materials.
7. The semiconductor structure of claim 5, further comprising a third barrier layer disposed between the first barrier layer and the conductive layer, wherein the third barrier layer comprises a fourth metal, the oxide barrier layer further comprises a sixth portion in direct contact with the third barrier layer, and the sixth portion comprises a fifth oxide of the fourth metal.
8. The semiconductor structure of claim 7, wherein the first barrier layer is a metal silicide layer, the second barrier layer is a metal layer or a first metal nitride layer, the third barrier layer is a second metal nitride layer, and the first metal nitride layer and the second metal nitride layer have different materials.
9. The semiconductor structure of claim 1, wherein the bit line capping layer comprises: a first nitride layer covering a second sidewall of the oxide barrier layer; an oxide layer covering a third sidewall of the first nitride layer; and a second nitride layer covering a fourth sidewall of the oxide layer and a top surface of the bit line structure.
10. The semiconductor structure of claim 1, wherein the conductive layer comprises a metal selected from the group consisting of W, Ru, Ir, Pt, Rh, and Mo.
11. The semiconductor structure of claim 1, wherein the bit line capping layer comprises a first nitride layer, an oxide layer, and a second nitride layer covering the oxide barrier layer in sequence,

and the first nitride layer and the oxide layer have flush top surfaces.

12. The semiconductor structure of claim 1, wherein the bit line capping layer comprises a first nitride layer, an oxide layer, and a second nitride layer covering the oxide barrier layer in sequence, and the second nitride layer is in contact with a top portion of the oxide barrier layer.

13. A method of manufacturing a semiconductor structure, comprising: receiving a bit line structure disposed on a substrate, wherein the bit line structure comprises: a conductive silicon layer; a conductive layer disposed on the conductive silicon layer, wherein the conductive layer comprises a first metal; and a hard mask layer disposed on the conductive layer, wherein the hard mask layer comprises an insulating material selected from the group consisting of Si₃N₄, SiCN, and SiC; performing a plasma treatment on the bit line structure to form an oxide barrier layer in direct contact with a sidewall of the bit line structure and a top surface of the hard mask layer by oxidizing a first surface portion of the conductive silicon layer, a second surface portion of the conductive layer, and a third surface portion of the hard mask layer; and forming a bit line capping layer covering the oxide barrier layer.

14. The method of claim 13, wherein the plasma treatment is performed at a temperature between 225° C. and 275° C., and a plasma gas is generated from an oxidizing gas and a reducing gas.

15. The method of claim 14, wherein a flow rate percentage of the reducing gas is between 30% and 50% in the plasma gas.

16. The method of claim 13, wherein a treatment time of the plasma treatment is between 20 seconds and 45 seconds.

17. The method of claim 13, wherein the bit line structure further comprises: a first barrier layer disposed between the conductive silicon layer and the conductive layer, wherein the first barrier layer comprises a second metal.

18. The method of claim 17, wherein performing the plasma treatment on the bit line structure further comprises: oxidizing a fourth surface portion of the first barrier layer.

19. The method of claim 13, wherein a plasma gas is generated from an oxidizing gas and a reducing gas, and the reducing gas comprises NH₃.
